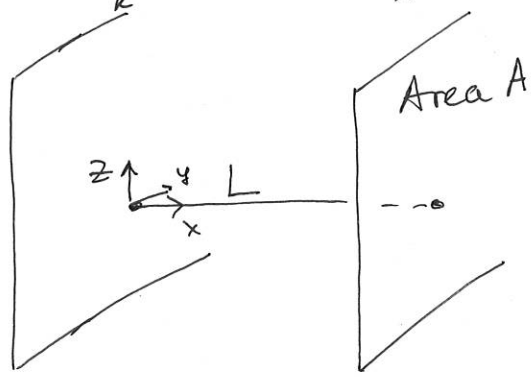
$T=0$, $\mathcal{E} = \frac{1}{2}\sum_{\vec{k}} E_{\vec{k}} = \frac{1}{2}\sum_{k}\sqrt{c^2\vec{k}^2+m^2}$



$|\vec{k}| = \sqrt{k_x^2+k_y^2+k_z^2}$

$k_x = \frac{\pi l}{L}$ due to the boundaries at $x=0$, $x=L$.

while $k_{y,z}$ are continuous.

Thus $\mathcal{E} = \mathcal{E}(L)$ due to k_x quantization

$$\mathcal{E}(L) = \frac{1}{2}A\sum_{l=0}^{\infty}\underbrace{\int dk_y\, dk_z}_{\int dq\, q\cdot 2\pi}\sqrt{c^2\frac{\pi^2 l^2}{L^2}+c^2\underbrace{(k_y^2+k_z^2)}_{q^2}+m^2}$$

$$= \frac{1}{2}\pi A\sum_{l=0}^{\infty}\int_0^{\infty} d(q^2)\sqrt{\frac{\pi^2c^2l^2}{L^2}+c^2q^2+m^2}$$

The force acting on the wall is $F = -\frac{\partial}{\partial L}\mathcal{E}(L)$

Regularization via "damping" factor $e^{-\eta\sqrt{\ }}$

$$\mathcal{E}(L) = \frac{1}{2c^2}\pi A\sum_{l=0}^{\infty}\int_0^{\infty} dx\sqrt{x+m^2+\frac{\pi^2c^2l^2}{L^2}}\, e^{-\eta\sqrt{\ }}$$

$$= \frac{\pi A}{2c^2}\sum_{l}\int_{z_l}^{\infty} dz\sqrt{z}\, e^{-\eta\sqrt{z}}$$

$z = x+m^2+\frac{\pi^2c^2l^2}{L^2}$, $z_l = m^2+\frac{\pi^2c^2l^2}{L^2}$ is the minimal value, for a given l.

Now $z = u^2$

$$\Rightarrow \mathcal{E}(L) = \frac{\pi A}{c^2}\sum_{l}\int_{\sqrt{z_l}}^{\infty} du\, u^2 e^{-\eta u}$$

But $u^2 e^{-\eta u} = \frac{\partial^2}{\partial \eta^2} e^{-\eta u}$

$$\mathcal{E}(L) = \frac{\pi A}{c^2} \frac{\partial^2}{\partial \eta^2} \sum_{l=0}^{\infty} \underbrace{\int_{\sqrt{z_l}}^{\infty} du\, e^{-\eta u}}_{\frac{e^{-\eta u}}{-\eta}\Big|_{\sqrt{z_l}}^{\infty} = \frac{1}{\eta} e^{-\eta\sqrt{z_l}}}$$

$$\mathcal{E}(L) = \frac{\pi A}{c^2} \frac{\partial^2}{\partial \eta^2} \sum_{l=0}^{\infty} \frac{1}{\eta} e^{-\eta\sqrt{m^2 + \frac{\pi^2 c^2 l^2}{L^2}}}$$

1) $l=0$ not important as it gives $\sim \frac{\partial^2}{\partial \eta^2}\left(\frac{1}{\eta} e^{-\eta m}\right)$

which is L-independent $\Rightarrow$ hence does not produce force!

2) let's set $m=0$ from now on.

$$\mathcal{E}(L) = \frac{\pi A}{c^2} \frac{\partial^2}{\partial \eta^2} \sum_{l=1}^{\infty} \frac{1}{\eta} e^{-\frac{\pi c l}{L}\eta}$$

but $\sum_{l=1}^{\infty} e^{-lb} = e^{-b} + e^{-2b} + e^{-3b} + \dots = e^{-b}(1 + e^{-b} + e^{-2b} + \dots)$

$$= \frac{e^{-b}}{1-e^{-b}} = \frac{1}{e^{b}-1}$$

$$\mathcal{E}(L) = \frac{\pi A}{c^2} \frac{\partial^2}{\partial \eta^2} \frac{1}{\eta} \frac{1}{e^{\frac{\pi c \eta}{L}} - 1}$$

Now expand in $\eta \to 0$ (at the end we need the result for $\eta = 0$)

$$\mathcal{E}(L) = \frac{\pi A}{c^2} \frac{\partial^2}{\partial \eta^2} \frac{1}{\eta} \left[\frac{1}{\frac{\pi c}{L}\eta} - \frac{1}{2} + \left(\frac{\pi c}{L}\right)\frac{\eta}{12} - \frac{1}{720}\left(\frac{\pi c}{L}\right)^3 \eta^3 \right]$$

$$F_1(L) = -\frac{\partial}{\partial L}\mathcal{E}(L) = -\frac{\pi A}{c^2}\frac{\partial^2}{\partial \eta^2}\frac{1}{\eta}\left\{\frac{\partial}{\partial L}\left(\frac{L}{\pi c \eta} - \frac{1}{2} + \frac{\pi c \eta}{12 L} - \right.\right.$$

$$\left.\left. - \frac{1}{720}\left(\frac{\pi c}{L}\right)^3 \eta^3\right)\right\} = \frac{\pi A}{c^2}\frac{\partial^2}{\partial \eta^2}\frac{1}{\eta}\left\{\frac{1}{\pi c \eta} - \frac{\pi c \eta}{12 L^2} + \right.$$

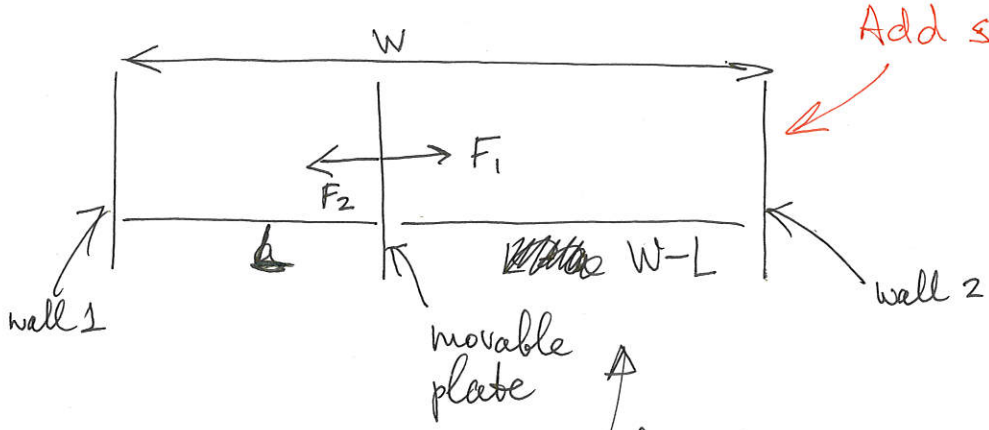
$$\left. + \frac{3}{720}\frac{\pi^3 c^3 \eta^3}{L^4}\right\} =$$

$$= -\frac{\pi A}{c^2}\frac{\partial^2}{\partial \eta^2}\left(\frac{1}{\pi c \eta^2} - \frac{\pi c}{12 L^2} + \frac{1}{240}\frac{\pi^3 c^3}{L^4}\eta^2\right)\Bigg|_{\eta=0}$$

infinite but
L-independent?

$$= -\frac{\pi A}{c^2}\frac{1}{120}\frac{\pi^3 c^3}{L^4}$$

Add second plate!



from here

$$F_2(L) = -\frac{\partial}{\partial L}\mathcal{E}(W-L) = -\frac{\pi A}{c^2}\frac{\partial^2}{\partial \eta^2}\frac{1}{\eta}\left\{\frac{\partial}{\partial L}\left(\frac{W-L}{\pi c \eta} - \frac{1}{2} + \frac{\pi c \eta}{12(W-L)}\right.\right.$$

$$\left.\left. - \frac{1}{720}\left(\frac{\pi c}{W-L}\right)^3 \eta^3\right\}$$

$$= -\frac{\pi A}{c^2}\frac{\partial^2}{\partial \eta^2}\frac{1}{\eta}\left\{\frac{-1}{\pi c \eta} + \frac{\pi c \eta}{12(W-L)^2} - \frac{1}{240}\frac{\pi^3 c^3}{(W-L)^4}\eta^2\right\}.$$

$$F_{TOTAL} = F_1 + F_2 = -\frac{\pi A}{120 c^2}\left\{\frac{\pi^3 c^3}{L^4} - \frac{\pi^3 c^3}{(W-L)^4}\right\} \xrightarrow[W\to\infty]{} -\frac{\pi^4 c A}{120 L^4}.$$